

Image Classification

Note: If you are using the factory-installed system image, there is no need to set up the environment from scratch; the environment is already configured. Simply follow the tutorial to execute the corresponding commands to utilize the features.

1. Image Classification: Single Image

Use the `yolo11n.pt` model to perform predictions on the sample images included within the Ultralytics project directory.

Navigate to the code directory:

```
cd /home/yahboom/ultralytics/ultralytics/yahboom_demo
```

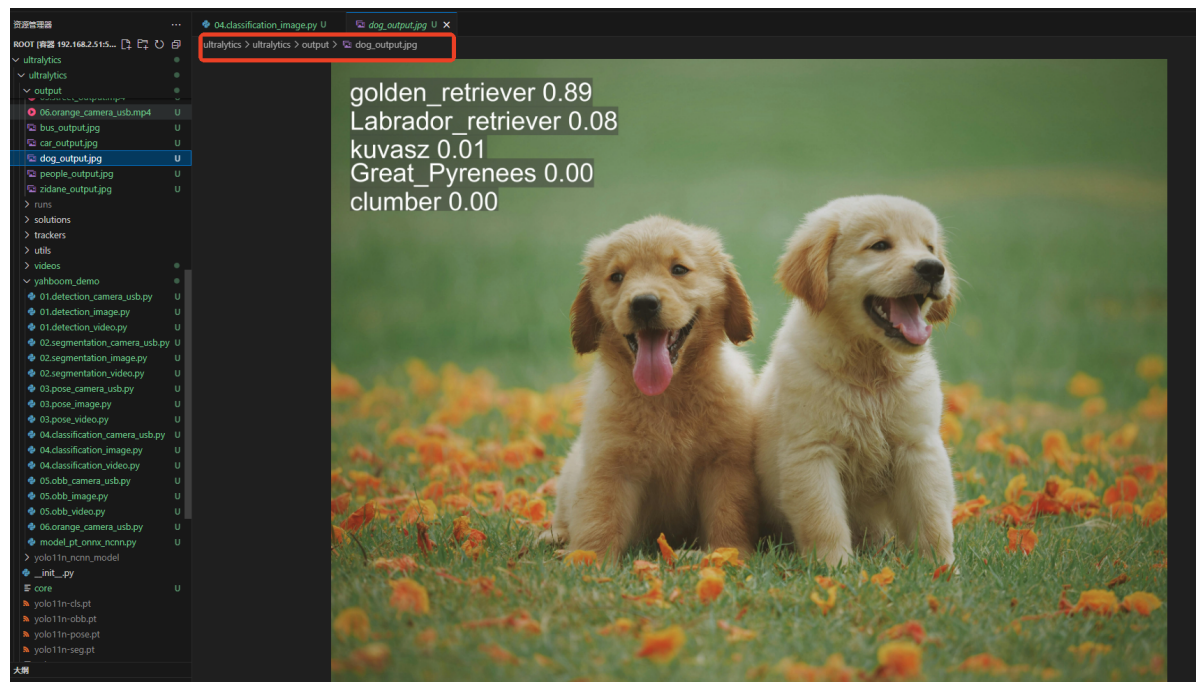
Run the code:

```
python3 04.classification_image.py
```

Preview of Results

Location of the output image generated by YOLO:

`/home/yahboom/ultralytics/ultralytics/output/`



Sample Code:

```
from ultralytics import YOLO

# Load a model
model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-cls.pt")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-cls.onnx")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-cls_ncnn_model")

# Run batched inference on a list of images
```

```

results = model("/home/yahboom/ultralytics/ultralytics/assets/dog.jpg") # return
a list of Results objects

# Process results list
for result in results:
    # boxes = result.boxes # Boxes object for bounding box outputs
    # masks = result.masks # Masks object for segmentation masks outputs
    # keypoints = result.keypoints # Keypoints object for pose outputs
    probs = result.probs # Probs object for classification outputs
    # obb = result.obb # Oriented boxes object for OBB outputs
    result.show() # display to screen

result.save(filename="/home/yahboom/ultralytics/ultralytics/output/dog_output.jp
g") # save to disk

```

2. Instance Segmentation: Video

Use `yolo11n.pt` to perform predictions on a video located within the Ultralytics project directory (specifically, a video *other than* the sample videos bundled with Ultralytics).

Navigate to the code directory:

```
cd /home/yahboom/ultralytics/ultralytics/yahboom_demo
```

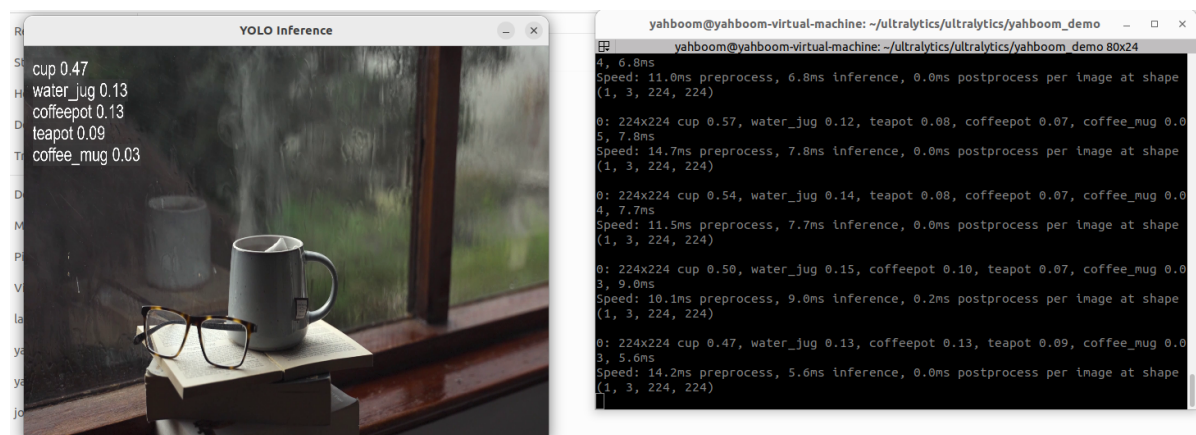
Run the code:

```
python3 04.classification_video.py
```

Preview of Results

Location of the output video (generated by YOLO detection):

`/home/yahboom/ultralytics/ultralytics/output/`



Sample Code:

```

import cv2
from ultralytics import YOLO

# Load the YOLO model
model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-cls.pt")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-cls.onnx")
# model = YOLO("/home/yahboom/ultralytics/ultralytics/yolo11n-cls_ncnn_model")

```

```

# Open the video file
video_path = "/home/yahboom/ultralalytics/ultralalytics/videos/cup.mp4"
cap = cv2.VideoCapture(video_path)

# Get the video frame size and frame rate
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = int(cap.get(cv2.CAP_PROP_FPS))

# Define the codec and create a Videowriter object to output the processed video
output_path = "/home/yahboom/ultralalytics/ultralalytics/output/04.cup_output.mp4"
fourcc = cv2.VideoWriter_fourcc(*'mp4v') # You can use 'XVID' or 'mp4v'
depending on your platform
out = cv2.VideoWriter(output_path, fourcc, fps, (frame_width, frame_height))

# Loop through the video frames
while cap.isOpened():
    # Read a frame from the video
    success, frame = cap.read()

    if success:
        # Run YOLO inference on the frame
        results = model(frame)

        # Visualize the results on the frame
        annotated_frame = results[0].plot()

        # Write the annotated frame to the output video file
        out.write(annotated_frame)

        # Display the annotated frame
        cv2.imshow("YOLO Inference", cv2.resize(annotated_frame, (640, 480)))

        # Break the loop if 'q' is pressed
        if cv2.waitKey(1) & 0xFF == ord("q"):
            break
    else:
        # Break the loop if the end of the video is reached
        break

# Release the video capture and writer objects, and close the display window
cap.release()
out.release()
cv2.destroyAllWindows()

```

3. Image Classification: Real-time Detection

3.1. Launching the Camera

Access the container terminal and navigate to the code directory:

```
cd /home/yahboom/ultralalytics/ultralalytics/yahboom_demo
```

Run the code: Click on the preview window, then press **Ctrl+C** to terminate the program!

```
python3 04.classification_camera_usb.py
```

Preview



Example Code Path:

/home/yahboom/ultralytics/ultralytics/yahboom_demo/04.classification_camera_usb.py